# SKILL – 4: Spring Dependency Injection – Constructor & Setter Injection

**#SKILL - 4 ➔ Spring Dependency Injection – Constructor & Setter Injection**

**Prerequisites:**

- **Basic knowledge of Java**
- **General Idea on Spring Framework**
- **Basic Idea on DI & IoC**

A training institute wants to automate the management of student information in a Spring application. The system should inject details such as **studentId**, **name**, **course**, and **academic year** into objects without manually assigning values. This Skill demonstrates **Constructor Injection** and **Setter Injection** using both **XML** and **Annotation** configurations.

**Tasks:**

1. Create a Java class named **Student** with fields: studentId, name, course, year.
2. Create a **Constructor** that accepts all fields and assigns values.
3. Create **Setter methods** for at least two fields (example: course, year).
4. Configure the Student bean using:
   a. **XML Configuration** - The student should take:
      - a **POJO class**
      - an **XML configuration file**
      - a **main class** to load the container
   b. **Annotation Configuration** - The student should take:
      - a **POJO class**
      - a **Java class with necessary annotations**
      - a **main class** to load the container
5. Create a Spring configuration file (XML or Java-based).
6. Write a main class to load the Spring IoC container.
7. Retrieve the Student bean and print the injected values.
8. **Push the Spring (core) project to GitHub.**

You can use both:

- applicationContext.xml (XML autowiring)
- Annotations (@Autowired, @Component, etc.)

Spring supports mixed configuration.

| Concept | Meaning | Spring Annotation |
|---|---|---|
| IoC | Spring controls object creation | (Internal) |
| DI | Injection of dependencies | `@Autowired` |
| Autowiring | Automatic DI process | `@Autowired` |
| Bean | Object managed by Spring | `@Component`, `@Service` |

## Create Project In Eclipse / STS

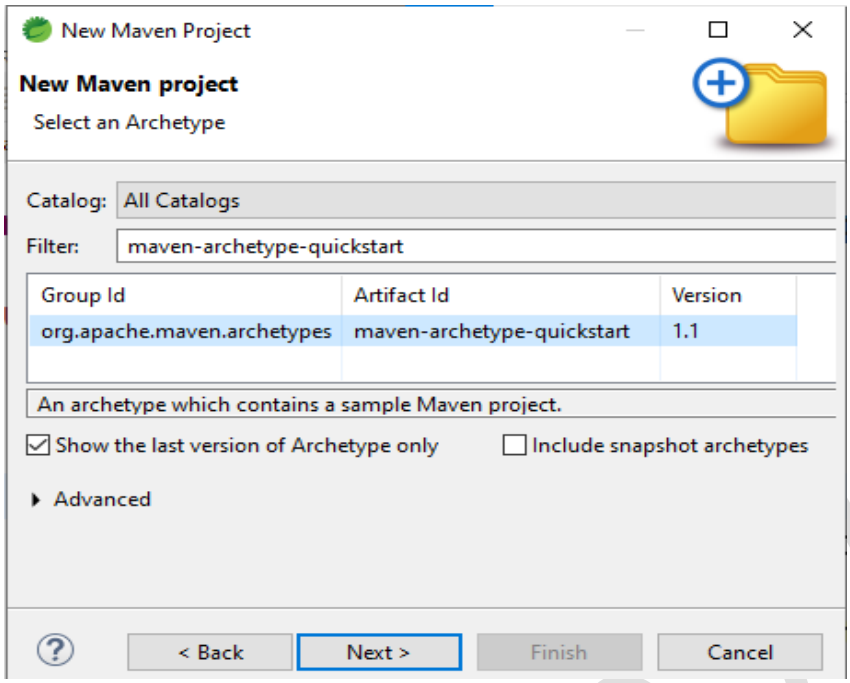
### File → New → Maven Project

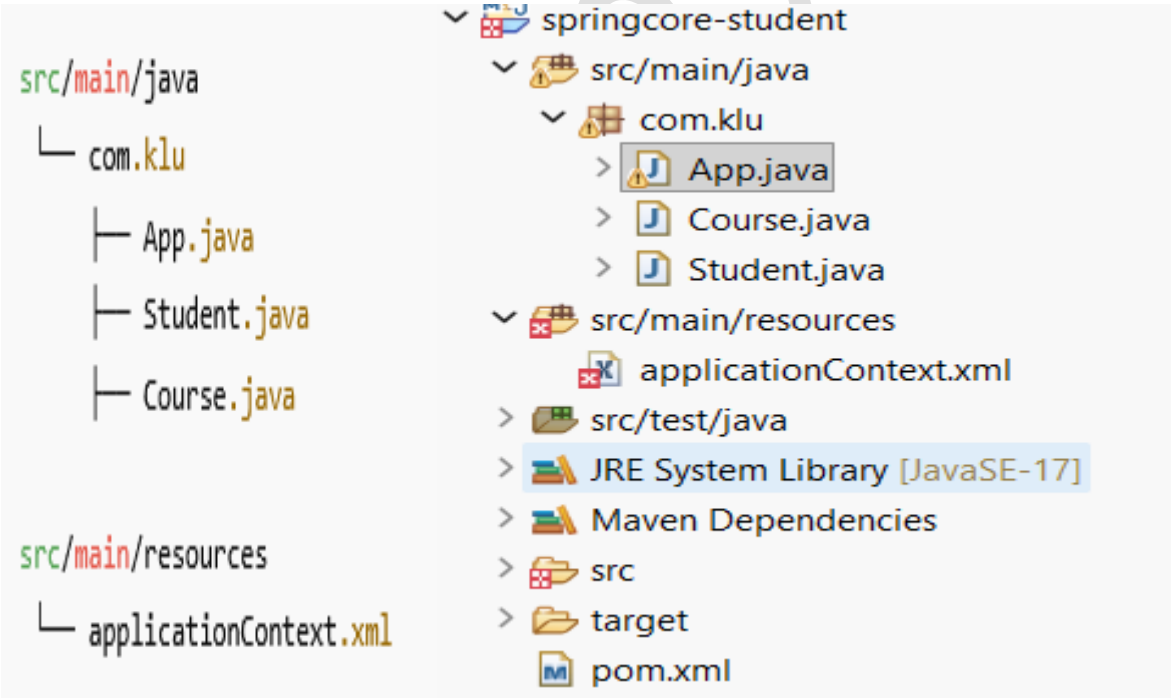
### ✓ Create a simple project

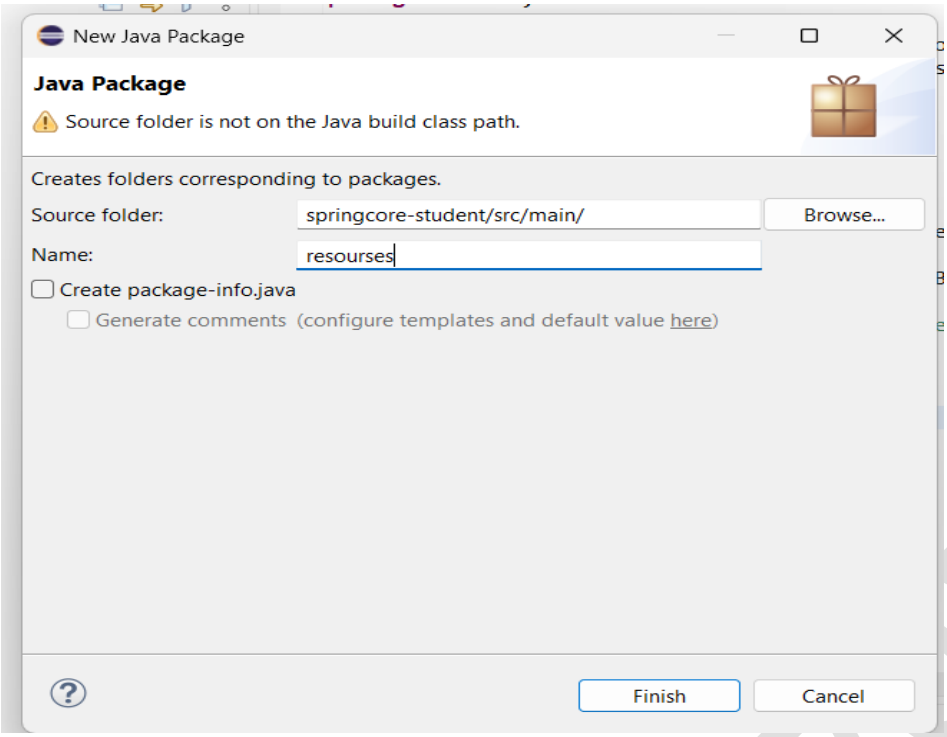
**Group Id : com.klu**

**Artifact Id : springcore-student**

**Packaging : com.klu**






## Add Spring Dependency (pom.xml):

---

```
<dependency>
<groupId>org.springframework</groupId>
<artifactId>spring-context</artifactId>
<version>5.3.30</version>
</dependency>
```

---

## Course.java :

```
package com.klu;
public class Course {
    private String courseName;
    private int duration;   // duration in months
    // Constructor Injection
    public Course(String courseName, int duration) {
        this.courseName = courseName;
 this.duration = duration;
    }
    // Getters (optional but good practice)
    public String getCourseName() {
        return courseName;
    }
    public int getDuration() {
        return duration;
```

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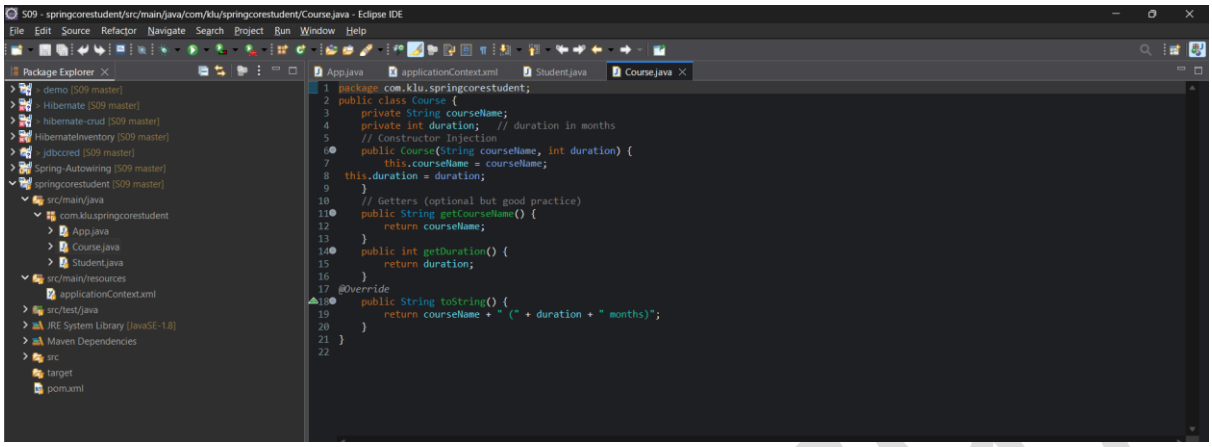
```
    }
@Override
    public String toString() {
        return courseName + " (" + duration + " months)";
    }
}
```




## Student.java:

```
package com.klu;
public class Student {
 private int studentId;
    private String name;
    private int year;
    private Course course;
   private String email;
    private String phoneNumber;
// Constructor Injection
    public Student(int studentId, String name, int year) {
        this.studentId = studentId;
        this.name = name;
        this.year = year;
    }
// Setter Injection
    public void setCourse(Course course) {
        this.course = course;
    }
public void setEmail(String email) {
        this.email = email;
    }
public void setPhoneNumber(String phoneNumber) {
        this.phoneNumber = phoneNumber;
    }
// Display method (each detail in a separate line)
    public void display() {
        System.out.println("Student ID      : " + studentId);
        System.out.println("Student Name    : " + name);
        System.out.println("Academic Year   : " + year);
```

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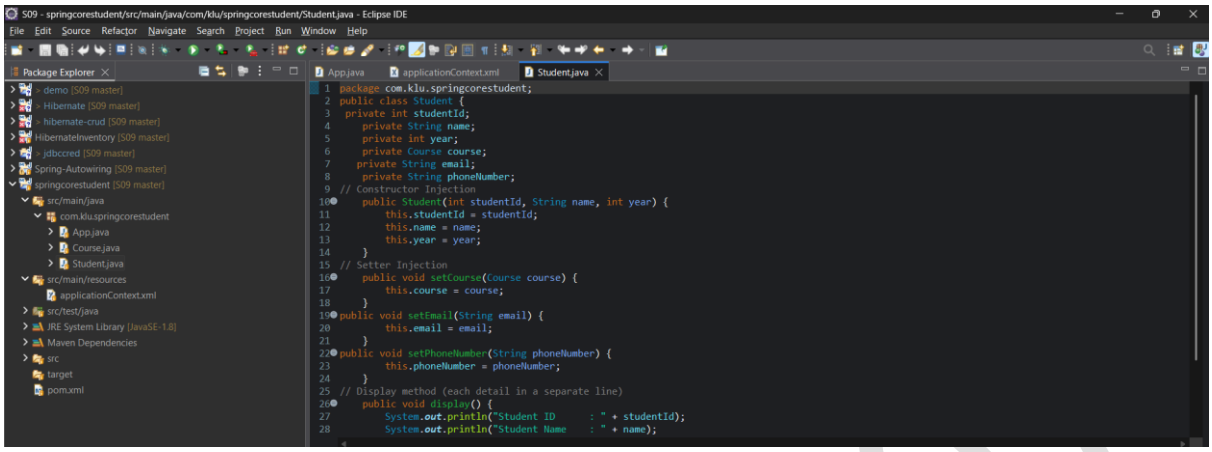
```
        System.out.println("Email          : " + email);
        System.out.println("Phone Number    : " + phoneNumber);
        System.out.println("Course         : " + course);
    }
}
```




## applicationContext.xml:

## 🗀 src/main/resources/applicationContext.xml

---

```
<?xml version="1.0" encoding="UTF-8"?>
<beans xmlns="http://www.springframework.org/schema/beans"
    xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
    xsi:schemaLocation="
    http://www.springframework.org/schema/beans
    https://www.springframework.org/schema/beans/spring-beans.xsd">

  <!-- Course Bean -->
  <bean id="course" class="com.klu.Course">
    <constructor-arg value="Spring Framework"/>
    <constructor-arg value="6"/>
  </bean>

  <!-- Student Bean -->
  <bean id="student" class="com.klu.Student">

    <!-- Constructor Injection -->
    <constructor-arg value="101"/>
    <constructor-arg value="ram"/>
    <constructor-arg value="2026"/>
<!-- Setter Injection -->
    <property name="course" ref="course"/>
    <property name="email" value="ram@klu.edu"/>
    <property name="phoneNumber" value="9876543210"/>
</bean>
</beans>
```

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```xml
<beans xmlns="http://www.springframework.org/schema/beans"
       xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
       xsi:schemaLocation="
       http://www.springframework.org/schema/beans
       https://www.springframework.org/schema/beans/spring-beans.xsd">

    <!-- Course Bean -->
    <bean id="course" class="com.klu.springcorestudent.Course">
        <constructor-arg value="Spring Framework"/>
        <constructor-arg value="6"/>
    </bean>

    <!-- Student Bean -->
    <bean id="student" class="com.klu.springcorestudent.Student">

        <!-- Constructor Injection -->
        <constructor-arg value="101"/>
        <constructor-arg value="ram"/>
        <constructor-arg value="2026"/>

        <!-- Setter Injection -->
        <property name="course" ref="course"/>
        <property name="email" value="ram@klu.edu"/>
        <property name="phoneNumber" value="9876543210"/>
```

---

## App.java:

```
package com.klu;
import org.springframework.context.ApplicationContext;
import org.springframework.context.support.ClassPathXmlApplicationContext;
public class App {
public static void main(String[] args) {
  ApplicationContext context =
            new ClassPathXmlApplicationContext("applicationContext.xml");
 Student student = (Student) context.getBean("student");
// IMPORTANT: call display(), do NOT use println(student)
      student.display();
   }
}
```

```java
package com.klu.springcorestudent;

import org.springframework.context.support.ClassPathXmlApplicationContext;

public class App {
    public static void main(String[] args) {

        ClassPathXmlApplicationContext context =
                new ClassPathXmlApplicationContext("applicationContext.xml");

        Student student = context.getBean("student", Student.class);
        student.display();

        context.close();   // Close the context
    }
}
```

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**Run the Application:**

1. **Right-click App.java**
2. **Run As → Java Application**

## OUTPUT on the console:

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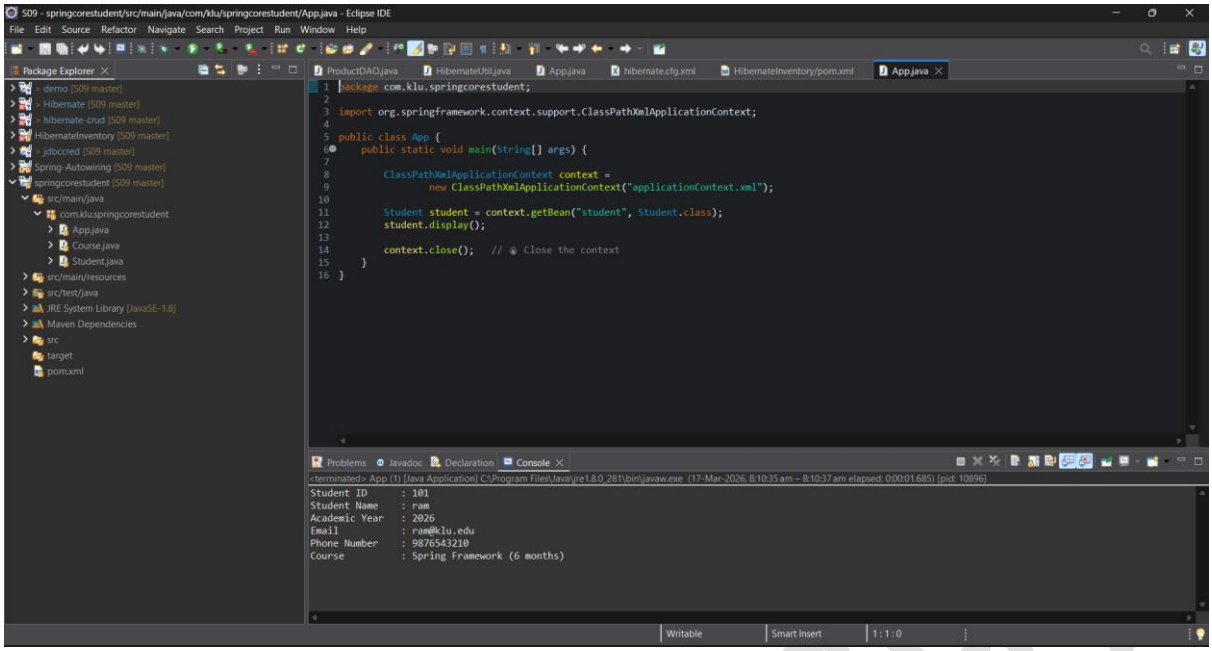
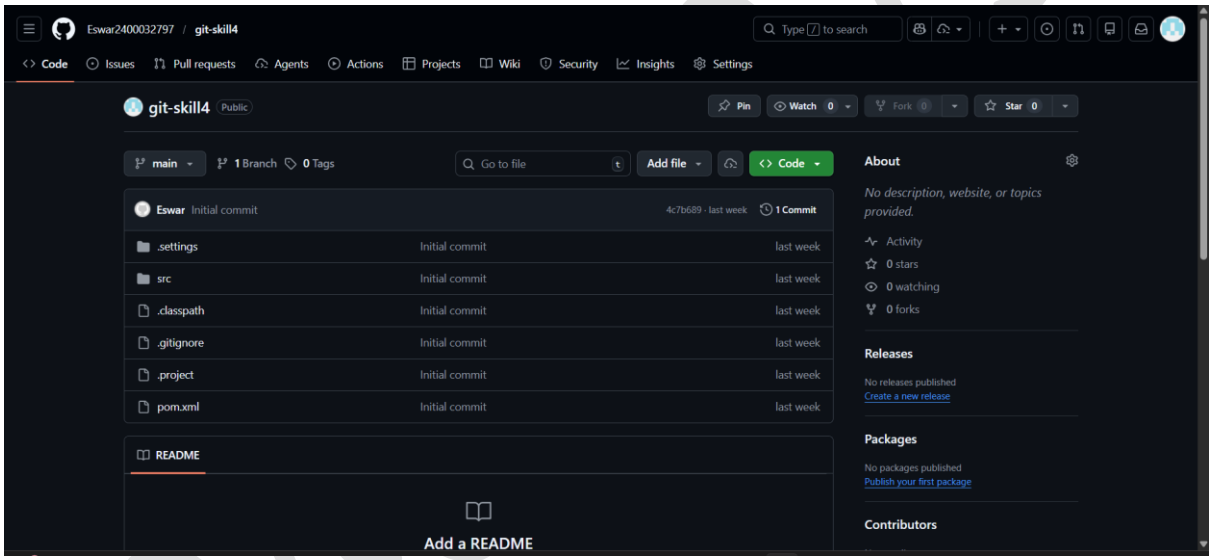



## Now Add the Springcore-students into the Github:




## GIT REPO LINK : https://github.com/Eswar2400032797/git-skill4.git